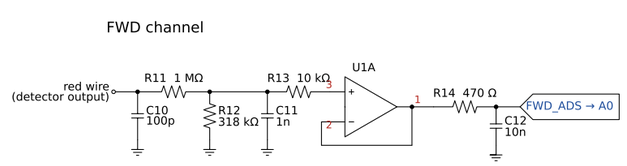
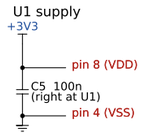
DigiMeter32 by SP2KMO — schematic diagram

ESP32-WROOM-32D (DevKit) · LCD 4" SPI (ILI9488/ILI9486/ST7796S) · MCP6002 · ADS1115 · all parts through-hole (THT), ready-made modules

A. INSIDE THE METER CASE — next to the coupler

1. Input buffer (U1 = MCP6002, DIP-8)





Divider R1/R2 = 1 MΩ / 318 kΩ (÷4.14);
R2 = 300 kΩ + 18 kΩ in series, R12 = R22.
Chosen from measurement (IC-7000, 40 m):
100 W → detector approx. 8.9 V
→ approx. 2.15 V at ADS1115
full range approx. 13 V (approx. 200 W)

Use the meter with FUNCTION = SWR —
the needle then does not load the detector
and the reading ignores RANGE/POWER/CAL.

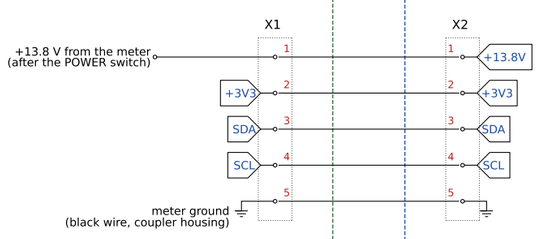
Solder C10/C20 right at the coupler.

The detector must give a POSITIVE voltage
to ground — check with a multimeter.

Inputs protected by R1 (1 MΩ) and the
diodes inside MCP6002 — extra Schottky
diodes intentionally omitted.

2. ADS1115 (module)

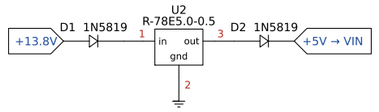
The module has 10 kΩ pull-ups
on SDA/SCL — do not add more.
Supply 3.3 V (not 5 V!) — then
inputs A0–A3 match the MCP6002
output (both on the same +3V3 rail,
brought by wire from the ESP).



Cable X1–X2: 5 wires (twisted or shielded),
up to approx. 50 cm. Shield grounded at the
meter side only. Analog FWD/REF signals
never leave the meter case.

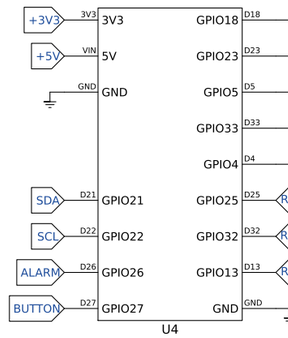
B. AT THE DISPLAY — on the back of the LCD panel

3. Power supply

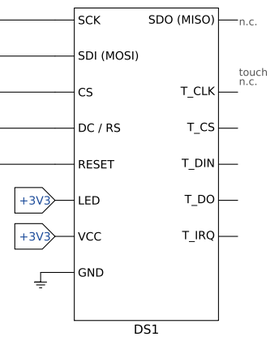


- U2 — ready-made converter (1 = +Vin, 2 = GND, 3 = +Vout, left to right facing the marking); no capacitors needed.
- D1 — reverse polarity protection.
- D2 — USB 5 V does not feed back into the converter; when working over USB, best set POWER = OFF.
- +3V3 for the LCD and (via X2) for MCP6002 and ADS1115 — from the 3V3 pin of the ESP32 board.
- Part B ground meets the meter only through connector pin 5 — do not tie it to the case separately.

4. ESP32-WROOM-32D (DevKit board)

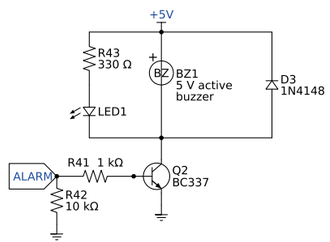


5. LCD 4" 480×320 SPI

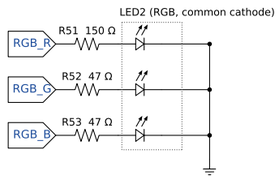


Pin order differs between modules —
wire by the NAMES printed on the board.
VCC and LED at 3.3 V work with any module.
LED tied high = backlight always on.

6. SWR alarm (optional)

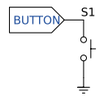


7. RGB status LED



LED legs (from left): 1 R, 2 K (common), 3 G, 4 B.
Green = standby, red = transmitting,
blue = ADS1115 missing.

8. Button



Momentary (NO), on D27.
Short: AVG/PEP,
long (>0.8 s): profile.

Pins as in the DigiMeter32 firmware (menuconfig, target esp32):
LCD SCK=D18, MOSI=D23, CS=D5, DC=D33, RST=D4 (VSPI), backlight always on;
I²C to ADS1115 SDA=D21, SCL=D22; button D27; alarm D26;
RGB LED R=D25, G=D32, B=D13.
Measurement via ADS1115 only — if it does not respond, the screen shows an ADS1115 error.
Nets with the same name (flags) are connected. THT parts, ready-made modules.

Revision history

Rev.	Date	Description	Author
1.0	2026-10-31	First release	SP2KMO
1.1	2026-11-04	Named DigiMeter32; R12/R22 318 kΩ; English	SP2KMO

Title	DigiMeter32 by SP2KMO — schematic diagram
Project	Digital power and SWR readout for an analog reflectometer (ESP32 + ADS1115 + LCD 4")
Drawn by	Kris SP2KMO
Date	2026-11-04
Revision	1.1
Sheet	1 / 1
Format	A3
File	docs/schematic.pdf (source: schematic.py)
Firmware	DigiMeter32, ESP-IDF 5.5, LVGL 9